Portfolio
Linkedin

Yaman Pandey (EIT)

(507) 304 5525
Yamanpandey2056@gmail.com

Education

- **Doctor of Philosophy (PhD) in Industrial and Systems Engineering** **2026 - 2030**
  Mississippi State University Starkville, MS
  Concentration: Manufacturing Systems
- **Master of Science in Manufacturing Engineering Technology** **2024 - May 2026**
  Minnesota State University, Mankato Mankato, MN
- **Bachelor of Science in Mechanical Engineering** **2018 - 2023**
  Minnesota State University, Mankato Mankato, MN

Work Experience

Reliability Engineer **dsm-firmenich** **Feb 2023 - Present**
New Ulm, MN

- Develop and execute comprehensive plant reliability strategy to improve equipment availability and reliability through the implementation of continuous improvement and defect elimination processes such as RCA (Root Cause Analysis) and RCM (Reliability Centered Maintenance).
- Interface with maintenance team to validate equipment performance and implement proactive maintenance practices while identifying and standardizing critical spare parts.
- Define PM and PdM procedures and frequency of implementation, optimize PM.
- Provide assistance with the Capex/Opex projects planning and execution.

Manufacturing Engineering Intern **Bombardier Recreational Products (BRP)** **2022**
St. Peter, MN

- Worked in the Alumacraft Boat Company under the marine group of BRP.
- Applied lean manufacturing principles to improve the operations and apply DMAIC methodology to identify root cause and implement permanent solution.
- Assisted on projects to improve assembly processes to meet cost and quality requirement.
- Designed/Assembled parts in CREO and Solidworks and performed FEA to upgrade various machines.
- Worked closely with Engineers, technicians, and labors to meet the goal of manufacturing 17 boats per day.
- Worked with Quality Engineers to ensure the product fulfils quality standers, meet safety regulations, and satisfy client requirement.

IT Tech **MNSU, Mankato** **2019-2022**

- Installed cameras, microphones, speakers, and Zoom PCs all over the campus classrooms and make a connection using an IP mixer to make sure they are all connected in a single system.
- Troubleshoot the problems in projectors, smartboards, and doc cam and fix them.
- Expert in creating, managing, escalating, and resolving Cherwell tickets by communicating with clients and staffs.
- Provide exceptional customer service which includes understandably communicating technical information, Provision network resources within Active Directory including printer access, admin access, and server access.
- Act as a liaison between end-users and management.
- Expert in creating, managing, escalating and resolving Cherwell tickets by communicating with clients and staff
  Utilize remote tools such as TeamViewer, Zoom, and VPN to solve issues remotely

Thesis

Vibration-Based Fault Diagnosis in Industrial Rotating Machinery
May 2026

- Analyzed vibration data from 32 rotating assets, including motors and fans, at a manufacturing site in Minnesota to support predictive maintenance.
- Applied data-driven and expert-validated vibration analysis to detect early indicators of rotating machinery faults and potential downtime risks.
- Supported reliability engineers in predicting required maintenance actions and reducing unplanned production downtime caused by rotary equipment failures.

Research

Integrating Smart Thermostat and Weather Forecasting Data for Real-Time HVAC Efficiency Optimization in Residential Buildings

2026

- Studied a forecast-informed thermostat control strategy for improving heating efficiency in a 2,256 ft² single-family residential building in Minnesota.
- Compared a conventional fixed setpoint baseline week with a forecast-adjusted thermostat week using smart Sensi thermostat and weather forecasting data.
- Found a 22.34% furnace runtime improvement at 0°F, showing that short-term weather forecasts can support practical and low-cost HVAC efficiency optimization.

Application of Lean Manufacturing in the Food Powder Products Sector

Jan 2024- April 2024

- Researched on the application of the lean manufacturing in the food powder products sector for a graduate level lean manufacturing class
- Determined how lean manufacturing is being used on the food powder plants.
- Investigated on the areas where the powder food industry could be improved in terms of lean principles.
- Wrote a research paper on the research findings and suggestions.

Skills

- Software: Minitab, R-Lab, MATLAB, Python, MS-Project, AutoCAD, BricsCAD, DraftSight, CREO, Simulate, SolidWorks, Simulink, Office 365, MSOffice, MS Excel
- Technical: Reliability Centered Maintenance (RCM) , Root Cause Analysis (RCA), Ishikawa (Cause and Effect/ Fishbone), ABS Methodology, FMEA, 5S, 6-Sigma, Lean Manufacturing, Continuous Improvement, DFMA, GD&T

Leaderships

Treasurer

American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE)

Mankato, MN (May 2022 - Dec 2022)

- Served as a treasurer in ASHRAE, Mankato Chapter to create a budget plan, report, and conduct fundraising activities.
- Planned events and work with guest speakers to plan for the timing of an event.
- Attended various conference organized by other chapters of ASHRAE.
- Conducted various workshops including the Revit workshop.

President

Nepalese Student Community (NeStCom)

Mankato, MN (May 2019 - May 2021)

- Served as the president for 150+ members and was responsible for conducting meetings each week.
- Acted as an in-charge to create budget report to be presented to the general members.
- Responsible for managing events such as estimating the number of guests, and gathering supplies)

Certifications

- Lean Six Sigma Greenbelt certification pending
- Engineer-In-Training(EIT)